Bio-Inspired Optimization of MPC and PD Controller Parameters for Quadruped Locomotion

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Abstract*—This paper presents a simulation-based framework for optimizing Model Predictive Control (MPC) tracking cost weights and joint-level proportional-derivative (PD) gains within an MPC–Whole Body Control architecture for the ANYmal D quadruped robot. Particle Swarm Optimization (PSO) and Moth-Flame Optimization (MFO) are used to tune four MPC parameters associated with normalized centroidal momentum and 24 PD gains corresponding to the robot joints. Both tuning problems are evaluated in a ROS-based simulation environment using the same fitness function, which combines execution time and trajectory-tracking error. Sixteen independent optimization runs are performed for each algorithm and tuning problem. MFO achieved lower mean final fitness values in both tuning problems and lower fitness dispersion for PD tuning. For MPC tuning, the mean final fitness values were 13.077 for MFO and 13.404 for PSO, with no statistically significant between-algorithm difference detected ($p = 0.0621$). For PD tuning, the Wilcoxon rank-sum test detected a statistically significant between-algorithm difference ($p = 0.0116$). Validation of the MFO-derived configurations included 16 completed runs per controller configuration and trajectory. On rectangular and triangular trajectories, respectively, the mean CTE values were 0.0588 m and 0.0655 m for MPC tuning, and 0.0674 m and 0.0625 m for PD tuning. The corresponding mean traversal times were 29.69 s and 20.15 s for MPC tuning, and 26.92 s and 22.38 s for PD tuning. These results demonstrate the potential of bio-inspired optimization for systematic controller tuning in simulated quadruped locomotion.

***Index Terms*—Quadruped Locomotion, Model Predictive Control, Whole-Body Control, Bio-Inspired Optimization.**

I. Introduction

The field of robotics has experienced significant advances in recent years, with quadruped robots emerging as versatile platforms capable of traversing complex terrains and performing tasks such as exploration, inspection, and environmental monitoring in challenging environments [1]. Due to their inherent mobility, stability, and adaptability, quadruped robots are particularly suitable for industrial applications, including offshore oil platforms [2], wind turbine inspection [3], and hazardous environments [4] where reliable autonomous operation is required.

Despite these advances, achieving robust, stable, and energy-efficient locomotion in industrial scenarios remains a major challenge. Quadruped robots must operate on uneven surfaces, narrow passages, slippery structures, and under external disturbances [5], while carrying sensors and other payloads required for inspection and monitoring tasks [6]. Their locomotion performance depends not only on the high-level motion references generated by the Model Predictive Control (MPC), but also on the joint-level feedback provided by the proportional-derivative (PD) controller. Although MPC can explicitly account for system dynamics, constraints, and future trajectory predictions [7], its effectiveness strongly depends on the tuning of weighting matrices, cost-function coefficients, and other planning parameters [8]. Likewise, the execution of the planned motion is highly sensitive to the selection of the PD gains or impedance coefficients [9], [10]. Inappropriate MPC parameters may produce poor trajectory tracking, oscillatory or unstable motion, reduced robustness, and excessive computational effort, whereas poorly tuned PD gains may cause excessive joint compliance or stiffness, torque peaks, oscillations, increased energy consumption, and actuator wear. Because quadruped locomotion is highly nonlinear, multivariable, and strongly coupled, the manual tuning of both MPC and PD parameters is time-consuming and rarely yields globally optimal solutions, particularly when multiple

conflicting performance objectives must be considered simultaneously [11].

To address this issue, several studies have investigated optimization-based methods for tuning controllers in legged robots. Conventional approaches, such as gradient-based optimization [12] and Bayesian optimization [13], have shown promising results under restricted conditions but often face difficulties when dealing with nonconvex optimization problems and high-dimensional search spaces. In contrast, bio-inspired optimization algorithms have attracted considerable attention due to their capability to perform global optimization without requiring analytical gradients. Algorithms such as Particle Swarm Optimization (PSO), Genetic Algorithms (GA), Grey Wolf Optimizer (GWO), and Moth-Flame Optimization (MFO) have been successfully applied to robotic controller tuning [14], path planning [15], and engineering optimization problems [16]. In quadruped robotics, recent works have explored the use of these techniques for gait optimization [17], locomotion adaptation, and obstacle avoidance inspection [18]. Nevertheless, most existing studies focus on gait generation, locomotion benchmarks, or general navigation problems, while limited attention has been given to the optimization of both high-level MPC parameters and joint-level PD gains.

In this context, this paper proposes a bio-inspired optimization methodology based on PSO and MFO for the sub-optimal tuning of two complementary parameter sets within the locomotion controller of the ANYmal D robot. The first set comprises four MPC tracking cost weights associated with the normalized centroidal linear momentum along the x, y, and z directions and the normalized centroidal angular momentum around the z-axis. The second set comprises the proportional and derivative gains of the 12 joint actuators. The objective is to determine controller parameters that improve locomotion performance by reducing execution time and trajectory-tracking error. Candidate solutions are evaluated in a realistic three-dimensional simulation environment using detailed robot models and sensor configurations. The proposed framework enables systematic and automated tuning at both the planning and joint-feedback levels, reducing reliance on expert knowledge and supporting the deployment of quadruped robots in industrial applications.

The main contributions of this work are threefold. First, it introduces a unified simulation-based framework for optimizing both high-level MPC tracking cost weights and joint-level PD feedback gains using bio-inspired algorithms. Second, it formulates a common performance-oriented objective function that combines execution time and trajectory-tracking error, enabling a consistent evaluation of both tuning problems. Third, it validates the optimized parameters within an MPC–WBC locomotion architecture using the ANYmal D robot model in realistic three-dimensional simulations, assessing their effects on trajectory-tracking performance and the generalization of the optimized parameters to a reference path not used during optimization.

II. Locomotion Framework

Quadruped robots with torque-controlled actuation commonly employ hierarchical locomotion frameworks that integrate predictive planning with whole-body motion execution [19], [20]. In these architectures, the high-level planner generates feasible locomotion references considering the robot dynamics and contact interactions, while a lower-level controller executes these references through joint torque commands. Such a structure enables efficient handling of the floating-base dynamics of legged systems while maintaining real-time control performance at the actuator level.

Fig. 1 presents the overall locomotion architecture employed in this work, which consists of two main layers: a nonlinear Model Predictive Control (MPC) planner and a Whole-Body Control (WBC) execution module. The MPC layer operates at 100 Hz and predicts the future evolution of the robot states over a finite horizon. This planner computes reference trajectories for the robot base motion, joint velocities, and contact interaction forces while explicitly considering dynamic constraints and terrain contacts. The generated trajectories are then transmitted to the execution layer, which operates at 400 Hz, ensuring stable and accurate trajectory tracking during locomotion.

For the MPC, the robot dynamics is represented by a Centroidal Dynamics (CD) model [21]. The state vector is defined as $\mathbf{x} = (\mathbf{h}_{com},\ \mathbf{q}_b,\ \mathbf{q}_j) \in \mathbb{R}^{12+n_j}$ and the input vector is $\mathbf{u} = (\mathbf{f}_{c_1},\ ...,\ \mathbf{f}_{c_{n_c}},\ \mathbf{v}_j) \in \mathbb{R}^{3n_c+n_j}$, where $\mathbf{h}_{com}$ is the centroidal momentum, $\mathbf{q}_b$ is base pose (position and orientation), $\mathbf{q}_j$ are the joint positions, $\mathbf{f}_{c_i}$ are the contact forces and $\mathbf{v}_j$ are the joint velocities. The equations of motion are given by

$$\dot{\mathbf{h}}_{com} = \begin{bmatrix} \sum_{i=1}^{n_c} \mathbf{f}_{c_i} + m\mathbf{g} \\ \sum_{i=1}^{n_c} \mathbf{r}_{com,c_i} \times \mathbf{f}_{c_i} \end{bmatrix} \tag{1}$$

$$\dot{\mathbf{q}}_b = \mathbf{A}_b^{-1} \left(\mathbf{h}_{com} - \mathbf{A}_j \dot{\mathbf{q}}_j \right) \tag{2}$$

$$\dot{\mathbf{q}}_j = \mathbf{v}_j, \tag{3}$$

where $\mathbf{r}_{com,c_i}$ denotes the position of the contact point c_i with respect to the center of mass, and $\mathbf{A}(\mathbf{q}) \in \mathbb{R}^{6\times(6+n_j)}$ is the centroidal momentum matrix (CMM) [21], which is constructed as a function of the system's full kinematic configuration and the multi-body inertias: $\mathbf{A}(\mathbf{q}) = [\mathbf{A}_b(\mathbf{q}) \ \ \mathbf{A}_j(\mathbf{q})]$.

The cost function for the MPC is defined by the quadratic cost related to tracking the reference states and inputs for the trajectory:

$$L(\mathbf{x}, \mathbf{u}, t) = \frac{1}{2} \left\| \mathbf{x} - \mathbf{x}_d \right\|_{\mathbf{Q}}^2 + \frac{1}{2} \left\| \mathbf{u} - \mathbf{u}_d \right\|_{\mathbf{R}}^2, \tag{4}$$

where $\mathbf{Q}$ and $\mathbf{R}$ are positive-definite weighting matrices, $\mathbf{x}_d$ is the reference state, and $\mathbf{u}_d$ is the reference input. The matrix $\mathbf{Q}$ determines the relative importance of tracking each component of the state vector. In this work, the four diagonal weights associated with the centroidal linear momentum along the x, y, and z directions, and the centroidal angular momentum around the z-axis, $\{q_{p_x}, q_{p_y}, q_{p_z}, q_{l_z}\}$, are optimized using a

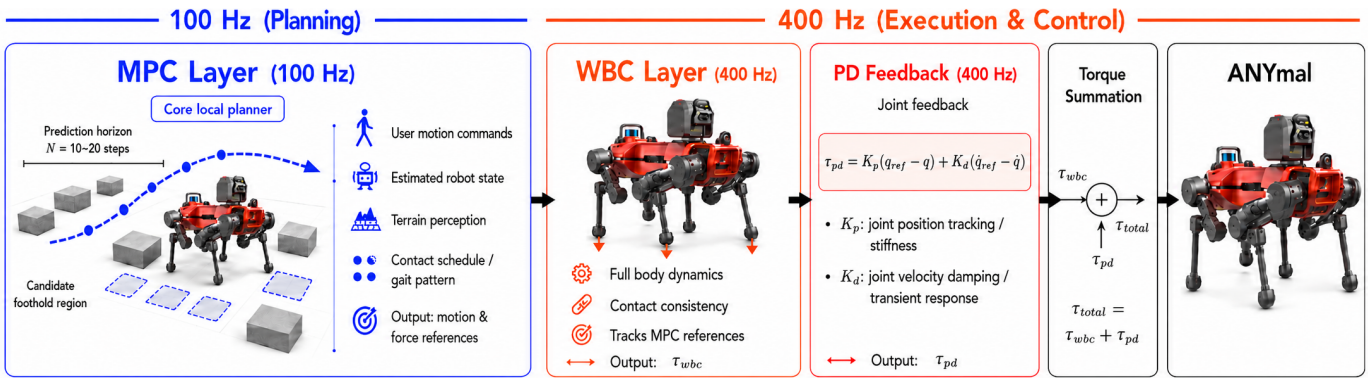


Fig. 1. Overview of the MPC–WBC–PD control architecture. The MPC generates motion references at 100 Hz. In the 400 Hz execution layer, the WBC torque and the joint-level PD feedback torque are summed to obtain the final joint torque commands for ANYmal.

bio-inspired algorithm, while the remaining entries are kept fixed. Larger values assign a higher penalty to deviations in the corresponding momentum component, thereby modifying the motion predicted by the MPC.

At the execution level, the WBC computes the desired torque vector by solving a constrained optimization problem that satisfies dynamic consistency, contact conditions, joint limits, and tracking objectives. Nevertheless, due to modeling uncertainties, discretization errors, actuator dynamics, terrain interaction effects, and state-estimation inaccuracies, tracking deviations may still arise during locomotion. To improve robustness and trajectory tracking performance, an additional feedback regulation term is incorporated into the execution layer, following commonly adopted strategies in torque-controlled legged robots [20], [22]. The resulting torques that are send to the joints are commanded by

$$\boldsymbol{\tau}_j = \boldsymbol{\tau}_{j,wbc} + \mathbf{K}_p \left(\mathbf{q}_j^{ref} - \mathbf{q}_j \right) + \mathbf{K}_d \left(\dot{\mathbf{q}}_j^{ref} - \dot{\mathbf{q}}_j \right), \quad (5)$$

where $\boldsymbol{\tau}_{j,wbc}$ is the torque computed by the WBC for the j-th joint, $\mathbf{q}_j^{ref}$ and $\dot{\mathbf{q}}_j^{ref}$ are the planned joint position and joint velocity, $\mathbf{q}_j$ and $\dot{\mathbf{q}}_j$ are the measured joint position and velocity, and $\mathbf{K}_p$ and $\mathbf{K}_d$ are the proportional and derivative gain matrices, respectively.

The optimized MPC tracking cost weights affect the generation of the whole-body motion references by determining how strongly deviations in centroidal momentum are penalized, thereby influencing base-motion tracking and the overall dynamic behavior of the robot. In contrast, the PD gains act locally at the joint level. $\mathbf{K}_p$ determines the tracking stiffness, whereas $\mathbf{K}_d$ provides damping and affects the transient response. Consequently, inadequate MPC weights mainly degrade the quality of the generated motion references, while poorly tuned PD gains may cause tracking errors, oscillations, or excessive torque commands during execution.

III. Parameter Tuning and Optimization

The following bio-inspired optimization algorithms were employed to determine sub-optimal controller parameters for the ANYmal D robot, including the MPC tracking cost weights and the joint-level PD gains. PSO, originally proposed by Kennedy and Eberhart [23], is a population-based metaheuristic inspired by the collective behavior of bird flocks and fish schools. Due to its simplicity, low computational requirements, and effective convergence properties, PSO has been widely applied to robotic control and optimization problems [24], [25].

The MFO algorithm was also considered. MFO is a swarm intelligence technique inspired by the navigation behavior of moths around light sources and was introduced by Mirjalili and Lewis [26]. Due to its strong global search capability, MFO has demonstrated competitive performance in controller tuning and robotic applications [27], [28]. In this work, PSO and MFO are used to optimize both the MPC tracking cost weights and the joint-level PD gains, and their performance is compared through statistical analysis.

This work addresses two complementary tuning problems in the locomotion controller of the ANYmal D robot, the optimization of the MPC tracking cost weights and the optimization of the joint-level PD feedback gains. Both problems are evaluated using the same simulation-based methodology, fitness function, and PSO and MFO implementations. The optimization is performed in an ROS-based environment using Gazebo and RViz, while the robot follows a square-shaped reference trajectory.

For the MPC, four tracking cost weights associated with the normalized centroidal momentum are optimized: *hcom_lin_x*, *hcom_lin_y*, *hcom_lin_z*, and *hcom_ang_z*. These parameters influence the base-velocity tracking performance and the overall dynamic behavior of the robot. For the joint-level PD controller, the proportional and derivative gains of the 12 actuators are optimized, resulting in a total of 24 parameters. The same fitness function is used in both cases. Therefore, only the decision variables, search-space dimension, and parameter bounds change between the MPC and PD tuning problems. The fitness function is defined as

$$\text{fitness} = w_t e_t + w_e \sum_{k=1}^{N} \sqrt{\begin{matrix}(x_k(\theta) - x_{\text{goal},k})^2 \\ + (y_k(\theta) - y_{\text{goal},k})^2\end{matrix}} \quad (6)$$

Here, θ denotes the set of parameters being optimized. For MPC tuning, it corresponds to the four tracking cost weights, whereas for PD tuning, it corresponds to the 24 proportional and derivative gains. The coefficients w_t and w_e weight the trajectory completion time e_t and the accumulated Euclidean tracking error, respectively. At each time step k, $(x_k(\theta), y_k(\theta))$ denotes the robot position obtained using the evaluated parameter set, whereas $(x_{\text{goal},k}, y_{\text{goal},k})$ denotes the corresponding reference position.

The weighting coefficients are set to $w_t = w_e = 0.5$, assigning equal importance to trajectory completion time and tracking accuracy.

For both tuning problems, PSO uses a population of seven particles, an inertia weight linearly decreasing from 0.9 to 0.1, cognitive and social coefficients of $c_1 = c_2 = 2.05$, and an initial particle velocity of 50. For the MPC tuning, the search space has four dimensions, and each parameter is bounded between 100 and 400. For the PD tuning, the search space has 24 dimensions, with lower and upper bounds of 10 and 100, respectively. MFO is configured using the same population size, search-space dimension, parameter bounds, maximum number of iterations, and stopping criteria adopted for PSO in each tuning problem.

Fig. 2 summarizes the proposed optimization methodology, illustrated for the MPC tuning problem. At each iteration, PSO and MFO generate candidate parameter sets, which are applied to the corresponding controller and evaluated in the ROS-based simulation environment. For MPC tuning, each candidate contains the four tracking cost weights, whereas for PD tuning, it contains the proportional and derivative gains of the 12 actuators.

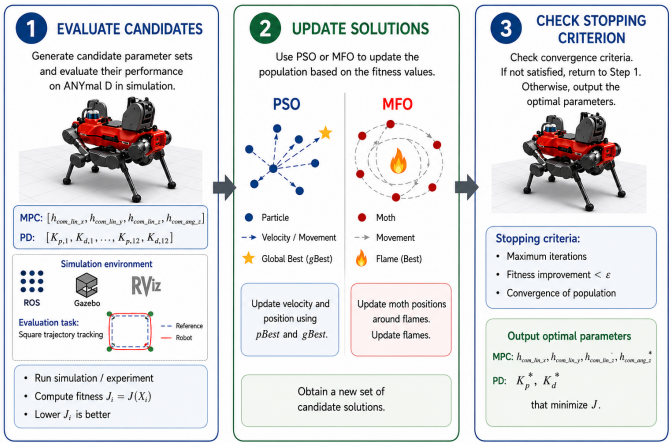


Fig. 2. Proposed MPC tracking cost weight optimization methodology using PSO and MFO. The same optimization loop is used for PD gain tuning by replacing the four MPC parameters with the 24 joint-level PD gains.

The robot then follows the square-shaped reference trajectory, and the completion time and accumulated trajectory error are used to calculate the fitness value in (6). This value is returned to the optimization algorithm, which updates its population according to the search mechanism of PSO or MFO. The process is repeated until the stopping criterion is satisfied, producing an optimized set of MPC tracking cost weights or joint-level PD gains.

To evaluate the repeatability of the optimization algorithms, each method is independently executed 16 times for each tuning problem. The resulting fitness values are first analyzed using the Shapiro–Wilk test at a 95% confidence level. If the normality assumption is satisfied, an Analysis of Variance (ANOVA) test is used to identify statistically significant differences between the algorithms. Otherwise, the Wilcoxon rank-sum test is applied.

IV. Results and Discussion

A. Controller Tuning Results

The proposed optimization framework was applied separately to two complementary tuning problems: MPC tracking cost weights and joint-level PD feedback gains. Tables I and II report the optimized parameters obtained with MFO and PSO and the final fitness statistics across 16 independent optimization runs per algorithm and problem. The algorithms produced distinct parameter configurations in both cases.

TABLE I
Optimized MPC parameters and statistical performance obtained using MFO and PSO.

Param.	MFO	PSO	Statistic	MFO	PSO
$hcom_lin_x$	150.00	258.81	Mean	13.077	13.404
$hcom_lin_y$	150.00	211.22	Median	13.050	13.264
$hcom_lin_z$	150.00	180.12	Std. Dev.	0.348	0.711
$hcom_ang_z$	400.00	246.35	Best	12.240	12.715

TABLE II
Optimized PD gains and statistical performance obtained using MFO and PSO.

Param.	MFO	PSO	Param.	MFO	PSO
P_1	100.00	29.70	D_1	68.90	25.20
P_2	10.00	24.20	D_2	31.11	44.04
P_3	41.47	78.24	D_3	64.84	31.15
P_4	90.99	60.91	D_4	83.20	46.45
P_5	87.78	21.83	D_5	32.55	11.60
P_6	24.67	51.65	D_6	98.68	40.92
P_7	97.80	37.28	D_7	10.00	55.48
P_8	10.00	12.37	D_8	24.95	65.04
P_9	74.21	83.56	D_9	53.11	62.99
P_{10}	100.00	40.93	D_{10}	70.40	62.69
P_{11}	21.33	69.95	D_{11}	24.80	44.95
P_{12}	15.86	99.96	D_{12}	65.31	29.13
Statistic	MFO		Statistic	PSO	
Mean	11.741		Mean	12.139	
Median	11.895		Median	12.057	
Std. Dev.	0.760		Std. Dev.	0.850	

For MPC tuning, Table I reports a lower mean final fitness for MFO than for PSO (13.077 versus 13.404), corresponding to a descriptive reduction of approximately 2.44%. The best fitness values were 12.240 and 12.715, respectively. The Wilcoxon rank-sum comparison yielded $p = 0.0621$; therefore, no statistically significant difference between the algorithms was detected at the 5% significance level.

For PD tuning, Table II reports lower mean (11.741 versus 12.139), median (11.895 versus 12.057), and standard deviation (0.760 versus 0.850) values for MFO than for PSO.

The Shapiro–Wilk tests rejected normality for both algorithms ($p = 0.000468$ for MFO and $p = 0.000176$ for PSO). The Wilcoxon rank-sum comparison yielded $p = 0.0116$, indicating a statistically significant between-algorithm difference in final fitness at the 5% significance level. These results supported selecting the MFO-derived PD gains for subsequent simulation experiments. Several MFO-optimized PD gains reached the search bounds.

B. Simulation Results

The MFO-derived MPC tracking cost weights and PD feedback gains were evaluated in simulation using the ANYmal quadruped robot. The rectangular path corresponds to the reference geometry used during tuning, whereas the triangular path provides an additional validation geometry. Figure 3 shows the reference paths and illustrative trajectories obtained with both controller configurations. The segments outside the reference paths correspond to the initial approach from the robot's initial position to the trajectory starting point, before tracking the prescribed reference path. The illustrated runs completed their prescribed waypoint sequences. The corresponding simulation videos are available in the supplementary material and on YouTube[1].

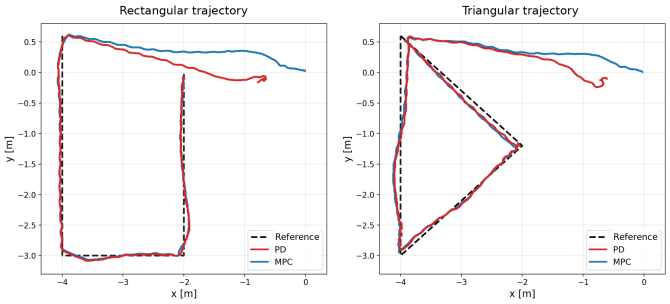


Fig. 3. Rectangular (left) and triangular (right) reference paths and illustrative trajectories obtained with the MPC-tuned and PD-tuned controllers.

Repeated validation provided 16 completed runs per controller configuration and trajectory. All retained runs were recorded as complete with five goals reached. Incomplete attempts were interrupted by crashes of the simulation environment and excluded from the tracking statistics. Table III reports completion counts over all recorded attempts and the mean and sample standard deviation of traversal time and per-run mean cross-track error (CTE). These averages are conditional on completion. CTE quantifies the deviation from the reference path, while traversal time is the sum of the recorded waypoint execution times.

As shown in Table III, MPC tuning yielded a lower mean CTE for the rectangular trajectory, whereas PD tuning yielded a shorter mean traversal time. For the triangular trajectory, PD tuning yielded a slightly lower mean CTE, while MPC tuning yielded a shorter mean traversal time.

[1]https://youtube.com/playlist?list=PLGNqdcsrdkI4&si=KY1prUsq79aHMoWs

TABLE III
REPEATED VALIDATION OF MFO-DERIVED PARAMETERS. VALUES ARE MEAN ± SAMPLE STANDARD DEVIATION ACROSS 16 COMPLETED RUNS. C/A DENOTES COMPLETED RUNS OVER RECORDED ATTEMPTS.

Path	Tuning	C/A	Time [s]	Mean CTE [m]
Rect.	MPC	16/18	29.69 ± 2.55	0.0588 ± 0.0586
Rect.	PD	16/25	26.92 ± 2.26	0.0674 ± 0.0259
Tri.	MPC	16/19	20.15 ± 4.11	0.0655 ± 0.0594
Tri.	PD	16/42	22.38 ± 4.06	0.0625 ± 0.0321

An exploratory comparison treated the completed runs as independent observations and used two-sided exact Mann–Whitney tests, with Holm adjustment across the four comparisons of CTE and time for the two trajectories. For the rectangular path, differences in ranks were detected for CTE ($p_{\mathrm{adj}} = 0.00023$) and time ($p_{\mathrm{adj}} = 0.0078$). For the triangular path, neither comparison was statistically significant at the 5% level after adjustment ($p_{\mathrm{adj}} = 0.5641$ for CTE and 0.0703 for time).

V. CONCLUSION

This paper presented a bio-inspired optimization framework for tuning two parameter sets in an MPC-WBC architecture for quadruped robot locomotion, namely MPC tracking cost weights and joint-level PD feedback gains. The proposed approach employed PSO and MFO with a common fitness function based on execution time and trajectory-tracking error.

The MPC-tuned and PD-tuned controllers differ in the parameter sets optimized at each layer. MPC tuning acts at the planning level by modifying the generated whole-body motion references, whereas PD tuning acts at the execution level by regulating joint-level tracking stiffness and damping. For the evaluated trajectories, the observed error and traversal-time profiles depended on the reference geometry and the metric considered.

For the triangular validation trajectory, the PD-tuned controller had a slightly lower mean CTE, whereas the MPC-tuned controller had a shorter mean traversal time. Neither rank-based comparison was statistically significant after Holm adjustment in the exploratory analysis. For the rectangular trajectory, the rank-based comparisons detected differences consistent with lower CTE values for MPC tuning and shorter traversal times for PD tuning. The PD-tuned controller also showed lower between-run dispersion of the mean CTE in the rectangular scenario. These observations refer to completed validation runs.

The statistical comparison between PSO and MFO depended on the tuning problem. PSO updates candidates using individual and collective information, whereas MFO uses moth–flame interactions. For MPC tuning, the Wilcoxon rank-sum test did not detect a statistically significant difference between the algorithms ($p = 0.0621$). For PD tuning, the test detected a statistically significant difference ($p = 0.0116$), and MFO had lower mean and median final fitness values. The trajectory validation presented here used only the MFO-derived parameter sets.

The similarity between objective-function summaries should be interpreted in terms of the selected fitness function. Since this function combines execution time and accumulated tracking error, similar fitness values can arise from different contributions of these terms. This is relevant to the nonlinear and coupled nature of quadruped locomotion. The optimization process therefore identifies candidate parameter configurations within the chosen search space, while the separate time and error metrics characterize their evaluated performance.

Neither PSO nor MFO guarantees a global optimum. Sixteen optimization runs per algorithm and problem characterized final-fitness variability, while repeated validation quantified MFO-derived tracking error and traversal time. Both algorithms use the same simulation-based evaluation loop.

Overall, MPC tuning modifies motion-reference generation and PD tuning joint-level feedback. These scenario-specific tracking-error and traversal-time assessments are limited to the tested simulation conditions and completed validation runs.

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